

Bayesian Statistical Analysis of Meteorological Data

Team 13

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Bayesian Statistics

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Dataset Overview

The dataset consists of **hourly meteorological measurements** with 26 columns:

- ▶ **Temperature & Humidity:** temperature_2m, relative_humidity_2m, dew_point_2m
- ▶ **Precipitation:** precipitation, rain, evapotranspiration (ET0)
- ▶ **Pressure:** pressure_msl, surface_pressure
- ▶ **Cloud Cover:** cloud_cover (%)
- ▶ **Solar Radiation:** shortwave, direct, diffuse radiation, DNI, GTI
- ▶ **Other:** vapour_pressure_deficit, wind speed/direction, is_day

Normal–Normal Model: Mean Unknown, Variance Known

We model daily soil temperature x_i (0–7 cm depth) as

$$x_i \mid \mu \sim \mathcal{N}(\mu, \sigma^2), \quad i = 1, \dots, n,$$

with $\sigma^2 = 0.0497$, $n = 9143$ and $\bar{x} = 0.1924$.

Priors & Posterior Two conjugate Normal priors were used:

Prior 1 (flat/Jeffreys-like): $\mu \sim \mathcal{N}(0, \sigma^2/\kappa_0)$ with $\kappa_0 = 10^{-6}$

Prior 2 (weakly informative): $\mu \sim \mathcal{N}(\bar{x}, \sigma^2/\kappa_0)$ with $\kappa_0 = 1$ The posterior is Normal:

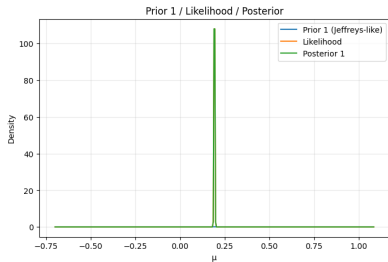
$$\mu \mid \mathbf{x} \sim \mathcal{N}\left(\mu_n, \frac{\sigma^2}{\kappa_n}\right), \quad \mu_n = \frac{\kappa_0 \mu_0 + n \bar{x}}{\kappa_0 + n}, \quad \kappa_n = \kappa_0 + n.$$

Results Due to the large n , both posteriors are identical:

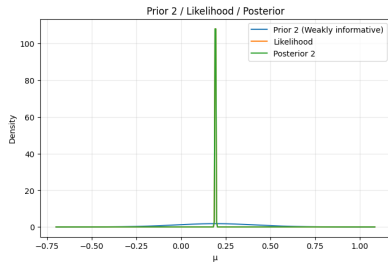
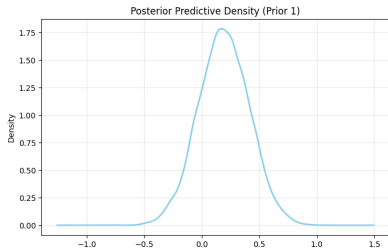
$$\hat{\mu}_{\text{post}} = 0.1924, \quad \text{SD} = 0.0023, \quad 95\% \text{ CI} = (0.1878, 0.1970).$$

Interpretation: Even with different priors, the data overwhelm the prior—both posteriors coincide. We are highly confident that the true mean soil temperature is about $(0.190 \pm 0.005)^\circ\text{C}$. **i.e** with lots of data, reasonable priors don't matter much—the data speak for themselves.

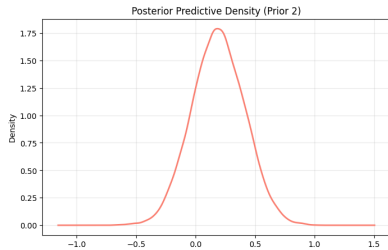
Posterior and Posterior Predictive density Comparison Under Two Different Priors



Prior 1: Flat ($\kappa_0 = 10^{-6}$)



Prior 2: Weakly informative ($\kappa_0 = 1$)



Dirichlet–Multinomial Model: Cloud Cover Categories

We categorize daily cloud cover ($N = 9144$ days):

Low ($\leq 20\%$): 3015 days

Medium (20%–80%): 1574 days

High ($> 80\%$): 4555 days

Let $\boldsymbol{\pi} = (\pi_1, \pi_2, \pi_3)$ be true category probabilities.

Posterior: $\boldsymbol{\pi} \mid \mathbf{n} \sim \text{Dirichlet}(\boldsymbol{\alpha}_0 + \mathbf{n})$.

Two priors tested:

Flat: $\text{Dirichlet}(1, 1, 1)$

Jeffreys: $\text{Dirichlet}(0.5, 0.5, 0.5)$

Posterior estimates (flat prior shown; Jeffreys differs < 0.001):

π_1 (Low): mean = 0.330, 95% CI = (0.319, 0.340)

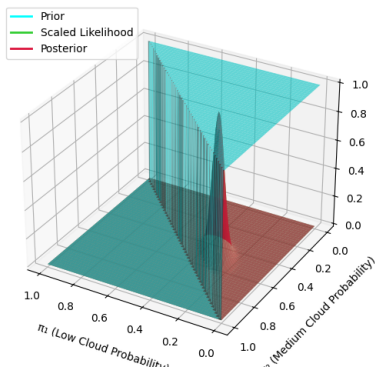
π_2 (Medium): mean = 0.172, 95% CI = (0.164, 0.180)

π_3 (High): mean = 0.498, 95% CI = (0.487, 0.509)

Interpretation: 50% of days are highly cloudy, 33% mostly clear, only 17% partly cloudy — a *bimodal* pattern (clear or overcast dominate).

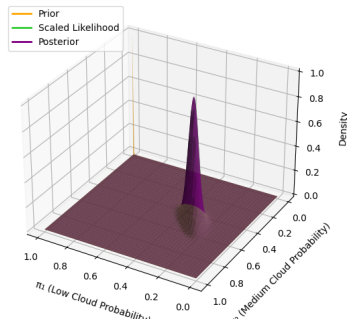
Posterior Comparison Under Two Priors

Flat Prior ($\alpha=[1,1,1]$) — Prior / Scaled Likelihood / Posterior



Cloud cover: Dirichlet(1,1,1)

Jeffreys Prior ($\alpha=[0.5,0.5,0.5]$) — Prior / Scaled Likelihood / Posterior



Cloud cover: Dirichlet(0.5,0.5,0.5)

Both models show nearly identical posteriors — with abundant data ($N = 9144$ for clouds), prior choice has negligible impact.

Poisson–Gamma Model: High-Radiation Event Counts

We model daily counts of high-radiation events ($\text{GTI} > 400 \text{ W/m}^2$) as

$$c_d \mid \lambda \sim \text{Poisson}(\lambda), \quad d = 1, \dots, D,$$

with $D = 381$ days and total events $\sum c_d = 2489$.

Priors on λ (rate per day):

Weakly informative: $\lambda \sim \text{Gamma}(1, 1)$

Jeffreys-like: $\lambda \sim \text{Gamma}(0.5, 0.1)$

Posterior: $\lambda \mid \mathbf{c} \sim \text{Gamma}(\alpha_0 + \sum c_d, \beta_0 + D)$.

Posterior estimates:

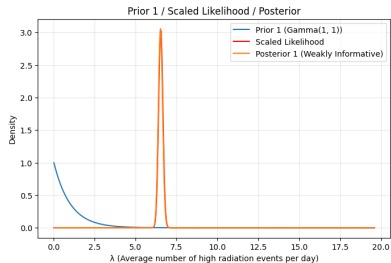
Weakly informative: $\lambda = 6.518$, 95% CI = (6.265, 6.777)

Jeffreys-like: $\lambda = 6.532$, 95% CI = (6.278, 6.791)

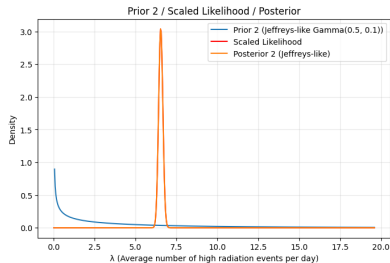
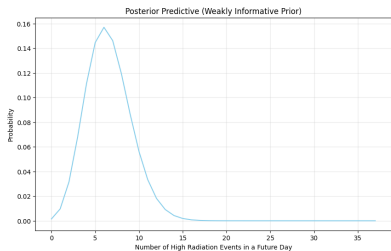
Interpretation: On a typical day, expect about (6.50 ± 0.25) high-radiation events. The two priors give nearly identical results — data dominate.

Takeaway: With sufficient data, reasonable priors don't change conclusions.

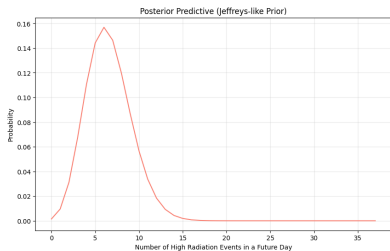
Posterior and Predictive Distributions: High-Radiation Events



Prior 1: $\lambda \sim \text{Gamma}(1, 1)$



Prior 2: $\lambda \sim \text{Gamma}(0.5, 0.1)$



Bayesian Hypothesis Testing: Is the Mean Temperature 25°C?

We tested whether the true mean of hourly temperature equals 25°C using Bayesian model comparison.

- ▶ Model: $T_i \sim N(\mu, \sigma^2)$
- ▶ Hypotheses: $H_0: \mu = 25^\circ\text{C}$ vs $H_1: \mu \neq 25^\circ\text{C}$
- ▶ Data: $\bar{T} = 26.488^\circ\text{C}$, $n = 9144$
- ▶ Prior: Weakly informative Normal–Inverse-Gamma prior on (μ, σ^2)
- ▶ Method: Exact Bayes factor via marginal likelihoods

Result: $\log \text{BF}_{10} = 8964.85 \Rightarrow \text{BF}_{10} \approx \infty$

Interpretation : “The Bayes factor is very large because the sample mean differs from the null value by 1.488°C and the dataset is very large ($n = 9144$). There is overwhelming evidence against H_0 .”

Grid Approximation – Conjugate Case (λ)

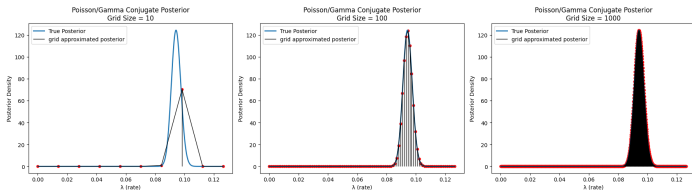
Model: $x_i \sim \text{Pois}(\lambda)$, $\lambda \sim \Gamma(\alpha_0, \beta_0)$ Posterior:

$\lambda \mid \text{data} \sim \Gamma(\alpha_0 + \sum x_i, \beta_0 + n)$

Grid comparison:

- ▶ 10 points: “very coarse and inaccurate”
- ▶ 100 points: “improves shape but distorts tails”
- ▶ 1000 points: “closely matches the true posterior”

Interpretation : “Grid-based Bayesian inference is reliable only when the grid is sufficiently fine.”



Grid Approximation – Non-Conjugate Case (k)

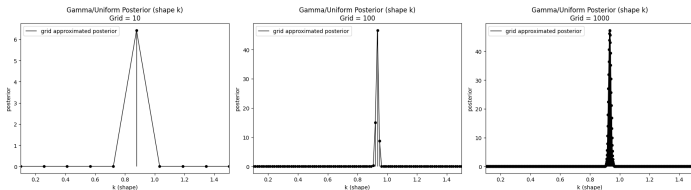
Model: $x_i \sim \Gamma(k, \theta)$, $\theta = \bar{x}$, $k \sim \text{Unif}(0.1, 1.5)$

“No conjugate prior for k ” $\rightarrow$ posterior computed numerically.

Grid results:

- ▶ 10 points: “crude, fails to capture true shape”
- ▶ 100 points: “much better but imperfect”
- ▶ 1000 points: “highly accurate, matches reference posterior”

Interpretation : “Dense grids can reliably handle non-conjugate Bayesian problems in 1D.”



Metropolis–Hastings for Shape Parameter k

Model: Wind speed $x_i \sim \Gamma(k, \theta)$,
 $\theta = \bar{x} = 15.4766$ fixed

Prior: $k \sim \text{Uniform}(0.1, 2.2)$

Algorithm:

— Propose $k^* = k^{(t)} + \varepsilon$,

$\varepsilon \sim N(0, 0.12^2)$

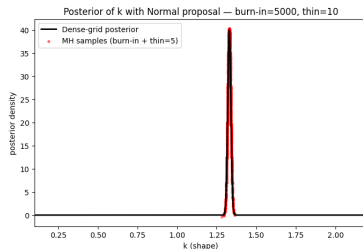
— Proposal is symmetric:

$q(k^*|k) = q(k|k^*)$, so acceptance ratio

simplifies to $\alpha = \min \left\{ 1, \frac{p(k^*|\text{data})}{p(k|\text{data})} \right\}$

Results:

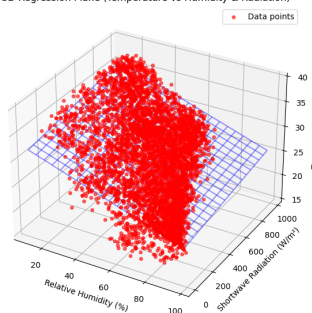
Posterior mean = 1.3296, sd =
0.0105, Acceptance rate = 0.106



Interpretation: The MH posterior closely matches the high-resolution grid reference, confirming accurate sampling. The narrow spread reflects strong data support for $k \approx 1.33$. Despite a modest acceptance rate, the 1D chain is stable and well-mixed.

Bayesian Multiple Linear Regression

3D Regression Plane (Temperature vs Humidity & Radiation)



Priors:

$$\beta \sim \mathcal{N}(\mathbf{0}, \sigma^2 \Lambda_0^{-1}), \Lambda_0 = 0.01 \mathbf{I}$$
$$\sigma^2 \sim \text{Inv-Gamma}(1, 1)$$

$$\text{Model: } T_i = \beta_0 + \beta_1 H_i + \beta_2 R_i + \varepsilon_i,$$
$$\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

(H : humidity, R : shortwave radiation)

Gibbs sampler (20k iters, 5k burn-in):

- ▶ $\beta_0 = 33.15$ (sd = 0.17)
- ▶ $\beta_1 = -0.1148$ (sd = 0.0021)
- ▶ $\beta_2 = 0.0037$ (sd = 0.0002)
- ▶ $\sigma^2 \approx 6.06$

Interpretation: The 3D regression plane shows that daytime temperature decreases with higher humidity (due to evaporative cooling) and increases slightly with shortwave radiation. The dense data points align well with the fitted surface, confirming a linear model provides a reasonable approximation within the observed range.